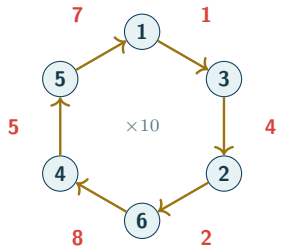
NUMBER THEORY LADDER 4 OF 9

Fermat, Euler and repeating decimals

Powers of a number modulo n cycle, and the cycle length explains both Fermat's theorem and the digits of $1/7$.

1 / THE RULE AND THE PICTURE

Long division of $1/7$ is the clock of remainders, multiplying by 10 each step



Each step turns the remainder r into the digit $\lfloor 10r/7 \rfloor$ and the new remainder $10r \bmod 7$, so the remainders run $1, 3, 2, 6, 4, 5$ and the digits $1, 4, 2, 8, 5, 7$ in red:

$$\frac{1}{7} = 0.\overline{142857}$$

The remainders are the powers $10^k \bmod 7$, and since they return to 1 after 6 steps the decimal repeats with period $6 = 7 - 1$.

2 / FERMAT'S LITTLE THEOREM

If p is prime and $p \nmid a$ then $a^{p-1} \equiv 1 \pmod p$

Multiplying the numbers $1, 2, \ldots, p-1$ by a only shuffles them modulo p, because a has an inverse. For $p = 7$ and $a = 3$ the list $1, 2, 3, 4, 5, 6$ becomes $3, 6, 2, 5, 1, 4$.

Taking products, $a^{p-1}(p-1)! \equiv (p-1)!$, and cancelling the unit $(p-1)!$ gives

$$\boxed{a^{p-1} \equiv 1 \pmod p}$$

Here $3^6 = 729 = 7 \cdot 104 + 1$, and Wilson's theorem $(p-1)! \equiv -1$ comes from pairing each number with its inverse, as in $6! = 720 \equiv -1 \pmod 7$.

3 / REPTEND LENGTHS / THE ORDER OF 10

The decimal of $1/p$ repeats with the smallest k for which $10^k \equiv 1 \pmod p$

p	$1/p$	period k	$p-1$	
3	$0.\overline{3}$	1	2	
7	$0.\overline{142857}$	6	6	full reptend
11	$0.\overline{09}$	2	10	
13	$0.\overline{076923}$	6	12	
17	$0.\overline{0588235294117647}$	16	16	full reptend
19	$0.\overline{052631578947368421}$	18	18	full reptend
23	$0.0434782608\ldots$ (22 digits repeat)	22	22	full reptend
29	$0.0344827586\ldots$ (28 digits repeat)	28	28	full reptend
31	$0.\overline{032258064516129}$	15	30	
37	$0.\overline{027}$	3	36	

The period always divides $p-1$, and primes where it equals $p-1$ are the full reptend primes, $7, 17, 19, 23, 29, 47, 59, 61, 97$ below 100, whose cycles of digits are the cyclic numbers.
For 11 the period 2 gives $0.\overline{09}$, and the primes 2 and 5 divide 10 and give terminating decimals.

4 / EULER'S THEOREM / COMPOSITE MODULI

If $\mathrm{HCF}(a, n) = 1$ then $a^{\varphi(n)} \equiv 1 \pmod n$, and the cycle length divides $\varphi(n)$

For $n = 10$ we have $\varphi(10) = 4$, and the last digits of the powers cycle: $3, 9, 7, 1$ for 3^k, $7, 9, 3, 1$ for 7^k, and $9, 1$ for 9^k, with cycle lengths $4, 4, 2$ all dividing 4. So $3^4 = 81$, $7^4 = 2401$ and $9^4 = 6561$ all end in 1, and the last digit of 7^{2023} is $7^{2023 \bmod 4} = 7^3 \equiv 3$. Fermat's theorem is the case $n = p$, where $\varphi(p) = p - 1$.

ASIDE / THE CYCLIC NUMBER

Multiplying 142857 by 1 to 6 only rotates its digits, and by 7 gives 999999.

LINKS TO OTHER SHEETS

The cycle of powers uses *Arithmetic on a clock*; the totient appears in *HCF, LCM, cicadas and the totient*.

Convention. — marks the repeating block; $\varphi(n)$ is Euler's totient; k is the multiplicative order of 10.